AI-Driven Fabric Defect Detection & Monitoring System for the Apparel Industry in SRI LANKA

Progress Presentation 1

25-26J-452

Faculty of Computing | Software Engineering

Sri Lanka Institute of Information Technology (SLIIT)

PROBLEM & IMPACT

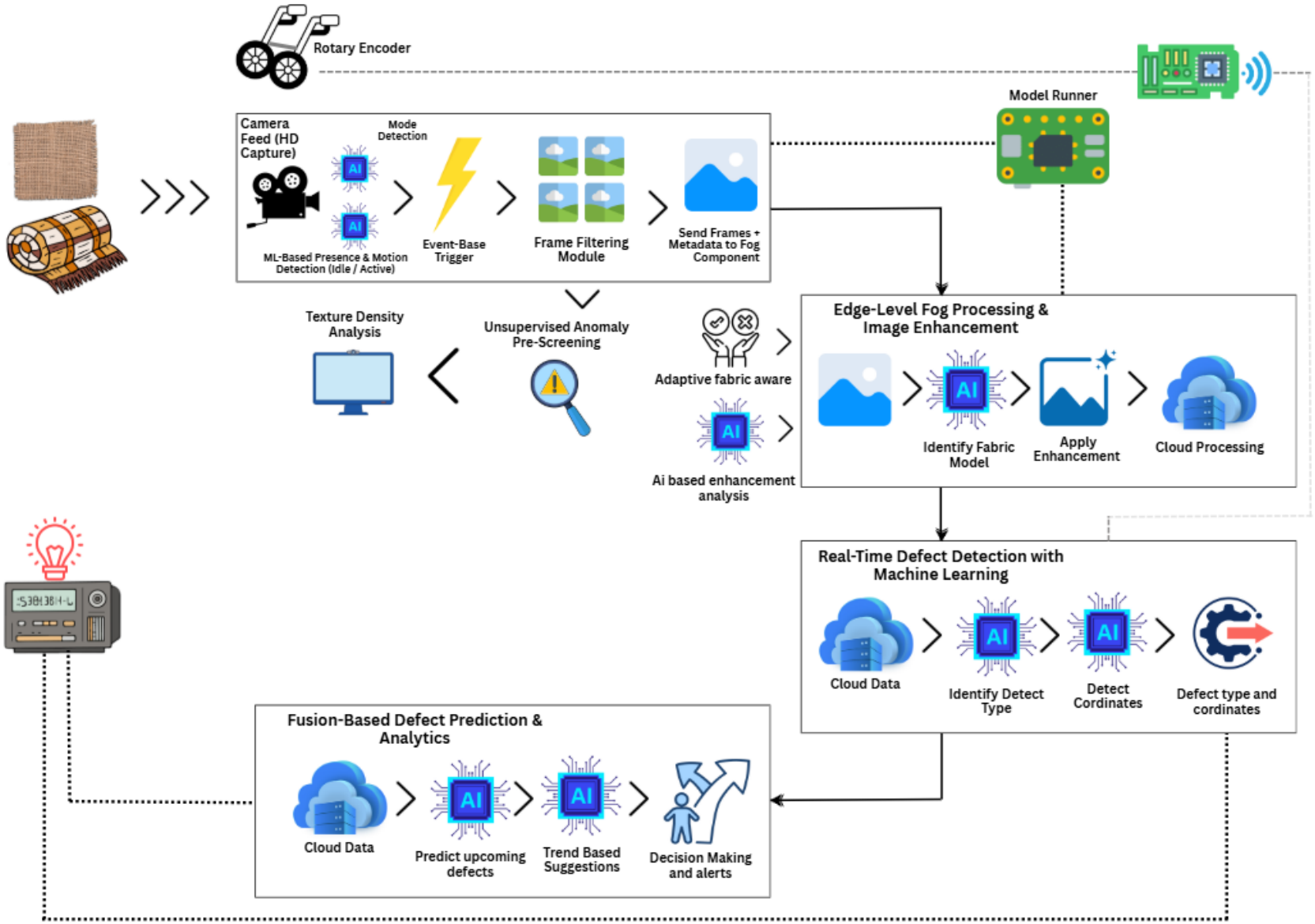
! Current Industry Problems

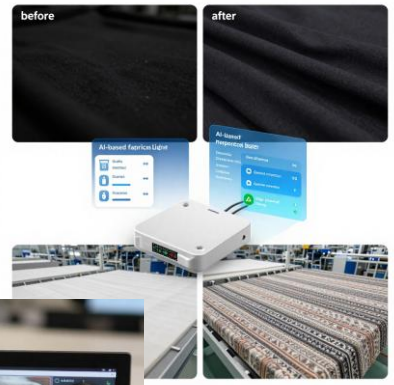
- *Fabric Inspection is manual*
- *Operators work at high speed*
- *Defects detected late*

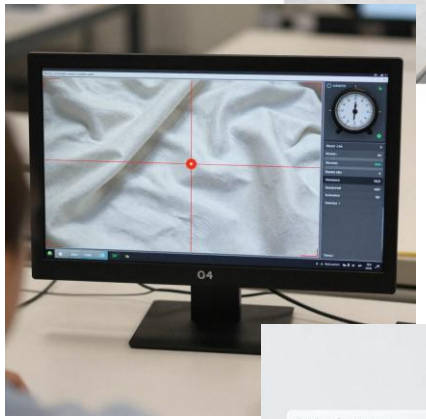
Result In

- *Fabric Wastage*
- *Higher Production Cost*
- *Quality Complaints & Rejections*

SYSTEM ARCHITECTURE







INTELLIGENT FABRIC CAPTURE, ANOMALY PRE-SCREENING, AND QUALITY ANALYTICS MODULE.

IT22573896 – Nagahawaththa J. C. D | 25-26J-452

Specialization:
SOFTWARE ENGINEERING

Sub Problem

- *Continuous fabric capturing generates huge volumes of redundant frames.*
- *Traditional systems capture every frame, Increasing latency and cost.*
- *No Intelligence at Capture Stage.*

User Requirements

- ***Reduce** unnecessary frame processing.*
- ***Real-time feedback** for operators.*
- ***Early detection** of abnormal fabric behavior & **forward only meaningful frames** to defect detection.*

Proposed Solutions

1. **Motion-Driven Frame Capture**
   (Frames captured only when fabric motion is detected)

2. **Unsupervised Anomaly Pre-Screening**
   (Autoencoder learns normal fabric behavior and flags only irregular frames)

3. **Frame-Level Quality & Routing**
   (Each suspicious frame is scored for quality and routed intelligently)

Design Excellence & Contribution

- *Designed an event-based, motion-aware capture using MobileNetV2, achieved ~98% accuracy.*
- *Developed unsupervised anomaly detection using AnomalyAutoencoder128 trained only on normal fabrics.*
- *Introduced Frame Irregularity Score (FIS) to quantify frame-level quality and enable intelligent routing.*

Prototype Validation & User Feedback

- *Models Validated using real fabric frames via live Python inference API.*
- *Designed and developed Operator friendly Dynamic Web dashboard using Next.JS.*
- *Implemented real-time alert system for better operator awareness.*

EDGE- LEVEL FOG PROCESSING & IMAGE ENHANCEMENT

Specialization:
SOFTWARE ENGINEERING

IT22574572 | Kamburugamuwa K.S.D | 25-26J-452

Sub Problem

- *Existing systems use fixed preprocessing, which cannot adapt to different fabric types or lighting conditions.*
- *Defect detection is often applied directly on unstable raw images at the edge.*
- Edge-level image quality stabilization is underexplored.

User Requirements

- Real-time, low-latency edge-level enhancement..
- Automatic adaptation to different fabric types.
- Forward only stabilized images for defect detection.

Proposed Edge-Level Solutions

- **ML-driven fabric-aware enhancement** that adapts settings in real time.
- **AI-based enhancement evaluation** using quality metrics and frame scores.
- **Operator dashboard** with visual analytics, alerts, and enhancement metrics.

Design Excellence & Contribution

- *Trained and evaluated **two ML models** for fabric-aware classification:*
  - *MobileNetV2 - **~ 87 Accuracy***
  - *EfficientNet - **~ 52 Accuracy***
- *Implement **real-time adaptive enhancement logic** based on ML predictions.*
- *Designed and implemented an **operator-facing web dashboard***
- Introduced **region-based fabric contribution analysis**

Prototype Validation & User Feedback

- ***Validated enhancement models using real fabric frames** captured from live production-like environments.*
- *Designed and developed Operator friendly Dynamic Web dashboard using Next.JS, Tailwind Based on user feedbacks.*
- Feedback influenced adaptive enhancement strategy selection.

REAL-TIME FABRIC DEFECT DETECTION & LOCALIZATION USING CV & IOT-BASED ROTARY ENCODER TRACKING

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Specialization:
SOFTWARE ENGINEERING

Problem Definition

- Current fabric quality control systems only detect defects and provide generic alerts without exact locations.
- Operators must manually search fabric rolls, resulting in time loss and reduced productivity.
- Manual inspection is prone to human error and inconsistency, leading to missed defects and quality variations.

Proposed Edge-Level Solutions

- ML-Driven Fabric defect detection with YOLOv9, detects defects like stains and holes.
- Encoder-Based Fabric Tracking: Custom rotary encoder with measuring wheel tracks fabric movement.
- Real-Time Defect Localization: Enables accurate location of defects on fabric rolls.

User Requirements

- Automatic detection and classification of fabric defects.
- Measure fabric length with exact defect positions using rotary encoder.
- Real-time visual alerts for immediate operator action.

Design Excellence & Contribution

- Trained and evaluated real-time fabric defect detection models:
  - YOLOv9 ~95% accuracy
  - CNN model ~70% accuracy
- Designed and implemented an operator-facing web dashboard using Next.js for real-time monitoring.
- Initiated the design of a rotary encoder–based measurement system to accurately track fabric dimensions during inspection.

Prototype Validation & User Feedback

- Validated fabric defect detection models using real fabric images captured from inspection scenarios via a live Python-based inference API.
- Designed and developed an operator-friendly dynamic web dashboard using Next.js to visualize detected fabric defects and inspection results, defect scheduling in real time.
- Designed defect scheduling functionality to automatically log detected defects and trigger notifications based on Cron-based tasks.

REAL-TIME FUTURE FABRIC DEFECT PREDCTION & PRESCRIPTIVE ANALYTICS MODULE FOR FUTURE FABRIC DEFECT RISK ASSESSMENT

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Specialization:
SOFTWARE ENGINEERING

Sub Problem

- Detect defects only after they occur, with no prediction of upcoming defect locations.
- Slow, error-prone, reducing proactive quality control.
- No risk scoring or root cause analysis to identify machine, material, or operator issues.

Proposed Solutions

- ML-Based Defect Prediction (LSTM): upcoming defect locations in real time.
- Predictive Risk & RCA (XGBoost and Random Forest): risk scores, defect pattern.
- Operator-Centric Insights: inspection zones, urgency levels, recommended actions.

User Requirements

- Ability to Predict future defect locations in real time.
- Provide risk scores and defect/root cause classification
- Support fast, edge-ready dashboard visualization.

Design Excellence & Contribution

- Trained and evaluated **two ML models** :
  - LSTM ~99.5% accuracy
  - XGBoost ~99.3% accuracy
- Engineered a FastAPI-based ML inference service for real-time predictions..
- Built a Next.js predictive analytics dashboard

Prototype Validation & User Feedback

- Validate models with unseen test data via a Python-based inference API
- Conducted iterative dashboard validation, incorporating user feedback to improve visualization

Commercialization

Customer Persona

- **Large & medium** garment / textile manufacturers.
- Quality managers, production supervisors, factory owners

Market Size

- Global textile & garment industry: USD 1+ trillion.
- Growing demand for automation, AI quality inspection

What is Unique in Our Solution

- **End-to-end pipeline**: capture → enhancement → defect detection → quality intelligence.
- **Prevents misleading defect analysis** using quality gating.

Cost Recovery / Revenue Model

- **Subscription (SaaS)** for analytics dashboard.
- Per-line / per-roll licensing.